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Method for producing electronic component mounted substrate

Abstract

A disclosed method for manufacturing an electronic-component-mounted substrate includes: a step (i) of forming a temporary fixing film (**13**) in such a manner as to cover a plurality of solder precoats (**11**) formed on a plurality of lands (**10b**) on a substrate and an antioxidation film (**12**) formed in such a manner as to cover the solder precoats; a step (ii) of disposing a plurality of electronic components (**30**) on the plurality of solder precoats (**11**) via the antioxidation film (**12**) and the temporary fixing film (**13**); and a step (iii) of soldering the plurality of electronic components (**30**) to the lands (**10b**) by melting the plurality of solder precoats (**11**). The antioxidation film (**12**) contains a first thermoplastic resin. The temporary fixing film (**13**) contains an activating agent and a second thermoplastic resin. The softening point of the first thermoplastic resin is equal to or lower than the softening point of the second thermoplastic resin.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a method for manufacturing an electronic-component-mounted substrate.

BACKGROUND ART

(2) A reflow method is known as a method for soldering a plurality of electronic components to a substrate. In the reflow method, electronic components are soldered with use of a solder precoat formed on the substrate in advance.

(3) PTL 1 (Japanese Laid-Open Patent Publication No. H6 (1994)-90079) proposes “a method for mounting electronic components on a printed circuit board including: printing a solder paste (8) on pads (3) in a rough pitch pad section (4) of a printed circuit board (1) including a fine pitch pad section (5) in which a pad arrangement pitch is less than 0.5 mm and the rough pitch pad section (4) in which the pad arrangement pitch is 0.5 mm or more; then applying a solder precipitating composition (12) containing an organic acid lead salt and tin powder as main components over the entire fine pitch pad section (5); then heating the printed circuit board (1) to form a solder layer (13) on the pads (3) in the rough pitch pad section (4) by melting the solder paste (8) and to form a solder layer (13) on pads (3) in the fine pitch pad section (5) by causing solder to precipitate from the solder precipitating composition (12); then applying a flux (15) to the printed circuit board (1); mounting electronic components (16) on the printed circuit board (1) and temporarily fixing leads (17) to the solder layers (13) on the pads (3) with adhesive power of the flux (15); and heating the printed circuit board (1), on which the electronic components (16) have been mounted, to solder the leads (17) of the electronic components (16) to the pads (3) by melting the solder layers (13).”

Paragraph 0026 of PTL 1 discloses that “it is preferable to clean the precoated substrate 14 with an alternative cleaning agent such as a surfactant-based cleaning agent, a higher alcohol-based solvent, or a terpene-based solvent, water, alternative CFC, or the like to remove a paste residue, a solder ball, and the like.”

CITATION LIST

Patent Literature

(4) PTL 1: Japanese Laid-Open Patent Publication No. H6 (1994)-90079

SUMMARY OF INVENTION

Technical Problem

(5) In recent years, miniaturization of electronic components is progressing, and a distance between

lands is reduced along with the miniaturization of electronic components. When a distance between lands is short, defects are likely to occur as a result of electronic components moving while being mounted.

(6) Under the above circumstances, an object of the present invention is to provide a method for manufacturing an electronic-component-mounted substrate that can suppress the occurrence of soldering defects even when a distance between lands is short.

Solution to Problem

(7) An aspect of the present invention relates to a method for manufacturing an electronic-component-mounted substrate. The method includes: a step (i) of forming a temporary fixing film in such a manner as to cover a plurality of solder precoats formed on a plurality of lands on a substrate and an antioxidation film formed in such a manner as to cover each of the plurality of solder precoats; a step (ii) of disposing a plurality of electronic components on the plurality of solder precoats via the antioxidation film and the temporary fixing film; and a step (iii) of soldering the plurality of electronic components to the plurality of lands by melting the plurality of solder precoats, wherein the antioxidation film contains a first thermoplastic resin, the temporary fixing film contains an activating agent and a second thermoplastic resin, and a softening point of the first thermoplastic resin is equal to or lower than a softening point of the second thermoplastic resin.

Advantageous Effects of Invention

(8) With the method for manufacturing an electronic-component-mounted substrate according to the present invention, it is possible to suppress the occurrence of defects when electronic components are soldered.

(9) Although novel features of the present invention are described in the appended claims, the following detailed description referring to the drawings will facilitate understanding of both the configuration and content of the present invention as well as other objects and features of the present invention.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1A is a cross-sectional view schematically showing a step in an example of a manufacturing method according to the present invention.

(2) FIG. 1B is a cross-sectional view schematically showing a step following the step shown in FIG. 1A.

(3) FIG. 1C is a cross-sectional view schematically showing a step following the step shown in FIG. 1B.

(4) FIG. 1D is a cross-sectional view schematically showing a step following the step shown in FIG. 1C.

(5) FIG. 1E is a cross-sectional view schematically showing a step following the step shown in FIG. 1D.

(6) FIG. 1F is a cross-sectional view schematically showing a step following the step shown in FIG. 1E.

(7) FIG. 1G is a cross-sectional view schematically showing a step following the step shown in FIG. 1F.

(8) FIG. 2 is a diagram schematically showing an example of temperature profiles of surfaces of a substrate in a step (iii).

(9) FIG. 3A is a cross-sectional view schematically showing a part of an example of a step of forming solder precoats.

(10) FIG. 3B is a cross-sectional view schematically showing a step following the step shown in FIG. 3A.

- (11) FIG. 3C is a cross-sectional view schematically showing a step following the step shown in FIG. 3B.
- (12) FIG. 3D is a cross-sectional view schematically showing a step following the step shown in FIG. 3C.
- (13) FIG. 3E is a cross-sectional view schematically showing a step following the step shown in FIG. 3D.
- (14) FIG. 4A is a cross-sectional view schematically showing a step in another example of the manufacturing method according to the present invention.
- (15) FIG. 4B is a cross-sectional view schematically showing a step following the step shown in FIG. 4A.
- (16) FIG. 4C is a cross-sectional view schematically showing a step following the step shown in FIG. 4B.
- (17) FIG. 4D is a cross-sectional view schematically showing a step following the step shown in FIG. 4C.
- (18) FIG. 4E is a cross-sectional view schematically showing a step following the step shown in FIG. 4D.
- (19) FIG. 4F is a cross-sectional view schematically showing a step following the step shown in FIG. 4E.
- (20) FIG. 4G is a cross-sectional view schematically showing a step following the step shown in FIG. 4F.
- (21) FIG. 4H is a cross-sectional view schematically showing a step following the step shown in FIG. 4G.
- (22) FIG. 5 is a top view schematically showing arrangement of lands and electronic components in Examples.

DESCRIPTION OF EMBODIMENTS

(23) The following describes examples of embodiments of the present invention, but the present invention is not limited to the following examples. In the present specification, the wording “from a numerical value A to a numerical value B” means a range including the numerical values A and B and can be read as “the numerical value A or more and the numerical value B or less”. In the following description, when examples of a lower limit value and examples of an upper limit value of a specific physical property, condition, or the like are described, any of the examples of the lower limit value and any of the examples of the upper limit value can be combined suitably unless the lower limit value is equal to or higher than the upper limit value.

(24) (Method for Manufacturing Electronic-Component-Mounted Substrate)

(25) A manufacturing method according to the present embodiment is a method for manufacturing an electronic-component-mounted substrate. Hereinafter, the manufacturing method may be referred to as a “manufacturing method (M)”. The manufacturing method (M) includes a step (i), a step (ii), and a step (iii) in this order.

(26) (Step (i))

(27) The step (i) is a step of forming a temporary fixing film in such a manner as to cover a plurality of solder precoats formed on a plurality of lands on a substrate and an antioxidation film formed in such a manner as to cover each of the solder precoats. The antioxidation film contains a first thermoplastic resin. The temporary fixing film contains an activating agent and a second thermoplastic resin. The softening point of the first thermoplastic resin is equal to or lower than the softening point of the second thermoplastic resin. The antioxidation film and the temporary fixing film contain the thermoplastic resins, and accordingly, these films soften when heated.

(28) The antioxidation film is formed in such a manner as to cover surfaces of the solder precoats. Therefore, the antioxidation film suppresses oxidation of the surfaces of the solder precoats. That is, the antioxidation film suppresses the occurrence of soldering defects due to oxidation of the surfaces of the solder precoats. It is possible to use a commercially-available substrate on which

solder precoats are formed, as the substrate including the solder precoats formed thereon.

Alternatively, the solder precoats may be formed on the lands of the substrate by using a method described later, for example.

(29) The first thermoplastic resin contained in the antioxidation film is rosin or a modified rosin, for example. In the following description, rosin and modified rosins may be collectively referred to as “rosins”. Examples of modified rosins include modified rosins described in Examples, but it is also possible to use other modified rosins, examples of which include natural rosins such as gum rosin and wood rosin, and derivatives thereof (e.g., a polymerized rosin, a hydrogenated rosin, a disproportionated rosin, an acid modified rosin, and a rosin ester). The softening point of a rosin can be changed by modifying the rosin. The antioxidation film may be a residue (a flux residue) formed on the surfaces of the solder precoats when the solder precoats are formed. In this case, a portion of a flux contained in a solder paste that is used to form the solder precoats forms the antioxidation film. In this case, a thermoplastic resin contained in the flux is the first thermoplastic resin contained in the antioxidation film. That is, a solder paste that contains the first thermoplastic resin is used in the case where a residue of the solder paste is used as the antioxidation film.

(30) The temporary fixing film temporarily and stably fixes electronic components disposed on the solder precoats until reflow is performed. The use of the temporary fixing film makes it possible to prevent a situation in which the electronic components are displaced and not mounted appropriately.

(31) The temporary fixing film can be formed by applying a temporary fixing agent (material of the temporary fixing film) in such a manner as to cover the solder precoats and the antioxidation film. After the temporary fixing agent is applied, the temporary fixing agent may be dried and/or subjected to heat treatment as necessary. There is no limitation on the method for applying the temporary fixing agent, and the temporary fixing agent may be applied through screen printing in which a mask is used, or may be applied with use of a dispenser.

(32) The temporary fixing film contains the activating agent and the second thermoplastic resin, and further contains other components as necessary. Rosins may be used as the second thermoplastic resin. Examples of the other components include a component that increases viscosity and a liquid component (a solvent or a dispersion medium). The activating agent is a substance that facilitates soldering. An activating agent that is used in a known solder flux may be used. Examples of the activating agent include abietic acid, succinic acid, glutaric acid, adipic acid, pimelic acid, suberic acid, azelaic acid, sebacic acid, dimer acid, propionic acid, 2,2-bis(hydroxymethyl)propionic acid, tartaric acid, malic acid, glycolic acid, diglycolic acid, thioglycolic acid, dithioglycolic acid, stearic acid, 12-hydroxystearic acid, palmitic acid, oleic acid, and modified acids of these acids. In addition to the above, examples of activating agents that have a reducing action include amines and halides. Examples of the component that increases viscosity include known components that are used as thickeners (or thixotropic agents), such as castor wax, an amide-based thixotropic agent, and a sorbitol-based thixotropic agent. The use of the temporary fixing film containing the activating agent makes it possible to suppress the occurrence of soldering defects by removing an oxide film from a terminal portion of an electronic component during the reflow step, for example.

(33) Examples of the liquid component contained in the temporary fixing film include hexyl diglycol (boiling point: about 260° C.). It is possible to use a liquid component that has a boiling point higher than the melting point of the solder constituting the solder precoats. For example, it is possible to use, as the liquid component, an alcohol, a polyol, a glycol ether, or any other organic compound (e.g., a glycol, a ketone, a hydrocarbon, an ester, or a terpeneol), each of which has a boiling point within a range from 230° C. to 320° C. Among these, substances that have a relatively high viscosity are preferably used.

(34) Examples of modified rosins that can be used as the second thermoplastic resin include the above-described examples of modified rosins that can be used for the antioxidation film. The

softening point of a rosin can be changed by modifying the rosin. The temporary fixing film may be formed using a solder flux that has a relatively high viscosity.

(35) The viscosity of the temporary fixing film may be within a range from 20 to 200 Pa.Math.s (e.g., 50 to 180 Pa.Math.s). When the temporary fixing film has a viscosity of 20 Pa.Math.s or more, the electronic components can be temporarily fixed stably. The viscosity of the temporary fixing film is particularly preferably 50 Pa.Math.s or more in order to temporarily fix the electronic components more stably. The viscosity can be measured using a rheometer manufactured by Anton Paar GmbH and a cone having an angle of 1.993°. Note that the viscosity described above is measured at 25° C.

(36) The softening point T1 (° C.) of the first thermoplastic resin is equal to or lower than the softening point T2 (° C.) of the second thermoplastic resin, and may be lower than T2. It can be considered that the antioxidation film starts to soften at the softening point T1 of the first thermoplastic resin. Likewise, it can be considered that the temporary fixing film starts to soften at the softening point T2 of the second thermoplastic resin. Therefore, the softening point T1 and the softening point T2 may be hereinafter described as the softening point of the antioxidation film and the softening point of the temporary fixing film, respectively. Note that softening points of rosins described in the present specification are values measured in accordance with a method described in JIS K 5902. Various rosins having different softening points are commercially available, and therefore, the softening points T1 and T2 can be easily adjusted using those rosins.

(37) The first thermoplastic resin is typically a main component of the antioxidation film. The second thermoplastic resin is typically a main component of the temporary fixing film. Here, the term “main component” refers to a component whose content is the highest among contents of components other than a liquid component (a solvent or a dispersion medium). The main component typically accounts for 50 mass % or more of the components other than the liquid component. Note that each of the first thermoplastic resin and the second thermoplastic resin may be constituted of a plurality of thermoplastic resins. In the case where the first thermoplastic resin and the second thermoplastic resin are each constituted of a plurality of resins, softening points of all the resins satisfy the relationship: softening point T1 softening point T2. However, the antioxidation film and the temporary fixing film may contain, as a trace component (whose content in the film is 5 mass % or less, for example), a thermoplastic resin other than the first thermoplastic resin and the second thermoplastic resin.

(38) The softening point T1 of the first thermoplastic resin may be within a range from 50° C. to 220° C. (e.g., 70° C. to 180° C.). The softening point T2 of the second thermoplastic resin may be within a range from 50° C. to 220° C. (e.g., 70° C. to 180° C.). The softening point T1 and the softening point T2 may satisfy a relationship: $0 \leq (T2 - T1) \leq 65$. The value (T2-T1) may be within a range from 0 to 110 (° C.). In the above ranges, the value (T2-T1) may be larger than 0. Note that, as a matter of course, the softening point T2 of the temporary fixing film is lower than the melting point of the solder precoat.

(39) In the case where the temporary fixing film contains a liquid component, it is preferable that both the first thermoplastic resin and the second thermoplastic resin dissolve in the liquid component of the temporary fixing film. In this case, the components of the antioxidation film and the temporary fixing film are miscible with each other when these films have softened. For example, it is preferable to use rosins as the first thermoplastic resin and the second thermoplastic resin and use a liquid component (e.g., hexyl diglycol) in which those rosins dissolve, as the liquid component of the temporary fixing film.

(40) (Step (ii))

(41) The step (ii) is a step of disposing a plurality of electronic components on the plurality of solder precoat via the antioxidation film and the temporary fixing film. There is no particular limitation on the method for disposing the electronic components, and a known method and a known device may be used. The plurality of electronic components may include an electronic

component having a size equal to or smaller than the 0402 size specified in JIS. The present invention is particularly effective when such a minute electronic component is mounted. The electronic components are fixed by the temporary fixing film, and thus stably fixed on the substrate until the step (iii) is performed.

(42) (Step (iii))

(43) The step (iii) is a step of soldering the plurality of electronic components to the plurality of lands by melting the plurality of solder precoats. The step (iii) can be performed as a step that is commonly called a reflow step. In the reflow step, the substrate and the like (the substrate on which the electronic components are disposed) are heated until the solder precoats melt, and then cooled to solidify the molten solder. Through the step (iii), the substrate on which the electronic components are mounted is obtained. Inspection on soldering or the like may be performed after the step (iii) as necessary.

(44) When the substrate on which the electronic components are disposed is heated in the step (iii), the antioxidation film softens first and thereafter the temporary fixing film softens. According to this configuration, it is possible to suppress the occurrence of a defect such as an electronic component being connected to a land other than a land to which the electronic component should be soldered, which defect occurs as a result of the electronic component moving while the substrate is heated. The shorter a distance between lands on which adjacent electronic components are disposed is, the higher the effect of suppressing the occurrence of a defect with this configuration becomes.

(45) There is no particular limitation on the step (iii) as long as the plurality of electronic components can be soldered to the substrate, and a known reflow step may be used. For example, the step of heating the substrate and the like may include a preheating step. Alternatively, a configuration is also possible in which the step of heating the substrate and the like does not include a preheating step. In such a case, the substrate and the like may be heated such that the temperature of an upper surface of the substrate increases at a temperature rise rate higher than $2^{\circ}\text{C./second}$ during a period for which the temperature of the upper surface of the substrate increases from 50°C. to the melting point of the solder precoats. That is, in this case, the temperature rise rate of the upper surface of the substrate does not become $2^{\circ}\text{C./second}$ or less during the period for which the temperature of the upper surface of the substrate increases from 50°C. to the melting point of the solder precoats. The temperature rise rate may be $3^{\circ}\text{C./second}$ or more, or $4^{\circ}\text{C./second}$ or more. The temperature rise rate may be $10^{\circ}\text{C./second}$ or less (e.g., $8^{\circ}\text{C./second}$ or less). When the temperature rise rate is $10^{\circ}\text{C./second}$ or less, separation of interconnects and lands from a substrate portion can be suppressed. Note that the upper surface of the substrate is a surface on which the lands for placing the electronic components thereon are formed. The time it takes to perform the step (iii) can be significantly reduced by omitting the preheating step and adopting a high temperature rise rate (e.g., $4^{\circ}\text{C./second}$ or more).

(46) When the solder precoats are exposed to high temperatures for a long period of time, surfaces of the solder precoats are oxidized and soldering defects are likely to occur. When the above-described temperature rise rate is $4^{\circ}\text{C./second}$ or more, oxidation of the surfaces of the solder precoats can be suppressed, and the occurrence of soldering defects can be suppressed. This effect is particularly noticeable when the plurality of electronic components include a minute electronic component. Here, the term “minute electronic component” refers to an electronic component having a planar shape in which the length of one side is 0.4 mm or less, for example. Such electronic components include electronic components having a size equal to or smaller than the 0402 size specified in JIS. In the present specification, the term “planar shape” refers to the shape of an electronic component as viewed from above the substrate in the state where the electronic component is mounted on the substrate.

(47) The step (iii) can be performed using a reflow apparatus having a configuration that is the same as or similar to the configuration of a known reflow apparatus. In order to suppress oxidation

of the surfaces of the solder precoat, the step (iii) may be performed in a nitrogen gas atmosphere in the reflow apparatus. However, even if the step (iii) is performed in an air atmosphere in the reflow apparatus, oxidation of the surfaces of the solder precoat can be suppressed in the case where the above-described temperature rise rate is 4° C./second or more. Accordingly, the step (iii) may be performed in an air atmosphere in the case where the above-described temperature rise rate is 4° C./second or more.

(48) The melting point of the solder precoat varies according to the solder that is used. In the case of a commonly-used lead-free solder (solder that does not contain lead), the melting point is 200° C. or higher, for example. Accordingly, when the lead-free solder is used, the temperature of the upper surface of the substrate is increased to 200° C. or higher (e.g., 220° C. or higher, or 230° C. or higher) in the step (iii). An upper limit of the temperature of the upper surface of the substrate may be set to 250° C. or lower (e.g., 240° C. or lower, or 230° C. or lower) in view of an influence on the electronic components.

(49) The shortest distance L between a first land on which one electronic component out of the plurality of electronic components is disposed and a second land on which an electronic component adjacent to the one electronic component is disposed may be 0.4 mm or less. The manufacturing method (M) is particularly effective when the shortest distance L is short. For example, the shortest distance L may be 0.03 mm or more, 0.05 mm or more, 0.1 mm or more, or 0.2 mm or more. The shortest distance L may be 0.4 mm or less, 0.3 mm or less, or 0.2 mm or less. However, the shortest distance L may be more than 0.4 mm. The shortest distance L may be within a range from 0.03 to 0.4 mm, a range from 0.05 to 0.4 mm, a range from 0.1 to 0.4 mm, or a range from 0.2 to 0.4 mm. The upper limit of these ranges may be changed to 0.2 mm or 0.3 mm unless the lower limit is equal to or higher than the upper limit.

(50) The distance between the first land and the second land may be various values in an electronic-component-mounted substrate, and the smallest value among those values is the shortest distance L. Assume a substrate that includes: a portion in which the distance between a first land on which one electronic component is disposed and a second land on which an electronic component adjacent to the one electronic component is disposed is 0.4 mm or less; and a portion in which the distance between a first land on which another electronic component is disposed and a second land on which an electronic component adjacent to the other electronic component is disposed is more than 0.4 mm. In the case of this substrate, the shortest distance L is 0.4 mm or less.

(51) The manufacturing method (M) may include a step (x) of forming the plurality of solder precoat and the antioxidation film on the plurality of lands before the step (i). There is no limitation on the method for forming the plurality of solder precoat, and a known method may be used.

(52) The manufacturing method (M) may further include, after the step (x) and before the step (i), a step (y) of pressing the plurality of solder precoat and the antioxidation film from above the substrate to form cracks in the antioxidation film and flatten upper surfaces of the plurality of solder precoat.

(53) The manufacturing method (M) may further include, after the step (x) and before the step (i), a step (z) of forming, in a case where at least one solder precoat has been formed inappropriately in the step (x) in such a manner as to short-circuit two adjacent lands, a notch in a portion of the inappropriately formed solder precoat, the portion being between and above the two lands.

(54) The manufacturing method (M) may include either the step (y) or the step (z) or both the steps (y) and (z). In the case where the both steps are included, the step (z) may be performed before or after the step (y).

(55) The following specifically describes examples of embodiments of the present invention with reference to the drawings. The following embodiments can be modified based on the above description. Also, matters described below may be applied to the above-described embodiment. Matters that are not essential to the embodiments of the present invention can be omitted. In the

drawings referred to below, the scale of members may be changed in order to facilitate understanding. Also, in the drawings referred to below, hatching may be omitted in some members. Note that, in the drawings referred to below, only one electronic component is shown as an electronic component mounted on a substrate, but a plurality of electronic components are actually mounted on the substrate.

Embodiment 1

(56) In Embodiment 1, an example of the manufacturing method (M) will be described. Steps included in the manufacturing method of Embodiment 1 are schematically shown in the cross-sectional views of FIGS. 1A and 1B.

(57) First, as shown in FIG. 1A, a temporary fixing film **13** is formed in such a manner as to cover each of a plurality of solder precoats **11** and an antioxidation film **12** (step (i)). The solder precoats **11** are formed on lands **10b** on a printed circuit board (substrate) **10**. The printed circuit board **10** includes a plate-shaped substrate portion **10a** and the lands **10b** formed on a surface of the substrate portion **10a**.

(58) Next, a plurality of electronic components **30** are disposed on the plurality of solder precoats **11** via the antioxidation film **12** and the temporary fixing film **13** as shown in FIG. 1B (step (ii)). Thus, a substrate **10X** on which the plurality of electronic components **30** are disposed is obtained. The illustrated example of the electronic components **30** includes two terminal portions **30a** and an element portion **30b** between the terminal portions **30a**.

(59) Next, the substrate **10X** is introduced into a reflow apparatus and heating of the substrate **10X** is started. The antioxidation film **12** softens earlier than the temporary fixing film **13**. FIG. 1C shows a state where the antioxidation film **12** has softened and the temporary fixing film **13** has not softened. At this time, the softened antioxidation film **12** spreads. However, the temporary fixing film **13** has not softened and maintains a shape corresponding to shapes of the solder precoats **11**. Therefore, even when the antioxidation film **12** softens, the electronic components **30** are kept from moving in a direction parallel to the surface of the printed circuit board **10**.

(60) Consequently, the occurrence of soldering defects is suppressed.

(61) FIG. 1D shows a state where the substrate **10X** has been further heated. In the state shown in FIG. 1D, the temporary fixing film **13** has also softened and the temporary fixing agent spreads along the surface of the electronic component **30**. As a result, an oxide film on surfaces of the terminal portions **30a** of the electronic component **30** is removed. Consequently, the occurrence of soldering defects is suppressed.

(62) FIG. 1E shows a state where the substrate **10X** has been further heated. In the state shown in FIG. 1E, portions of the solder precoats **11** have melted and molten cores **11b** are formed on the lands **10b**. Also, the components of the antioxidation film **12** and the temporary fixing film **13** are mixed with each other to be a mixture **13a**.

(63) When the substrate **10X** is further heated, the whole solder precoats **11** melt and form molten solder portions **11c** as shown in FIG. 1F. Thereafter, the substrate **10X** is cooled, and thus the molten solder is solidified and solder portions **11d** are formed as shown in FIG. 1G. The electronic components **30** are soldered to the lands **10b** via the solder portions **11d**. The temporary fixing film **13** becomes a residue **13g**. Thus, a substrate **10Y** on which the electronic components **30** are mounted is obtained. The substrate **10Y** is an electronic-component-mounted substrate.

(64) As described above, the substrate **10X** may be heated such that the temperature of an upper surface **10sa** of the printed circuit board **10** is lower than the temperature of a lower surface **10sb** of the printed circuit board **10**. Also, the substrate **10X** may be heated using either a method including the preheating step or a method not including the preheating step, as described above. Regarding the case where the preheating step is not performed, FIG. 2 shows an example of temperature profiles of the upper surface **10sa** and the lower surface **10sb** of the printed circuit board **10** in the step (iii).

(65) FIG. 2 shows an example in which the temperature of each surface of the substrate is increased

from room temperature (25° C.) to the highest temperature at a constant temperature rise rate. In the example shown in FIG. 2, the temperature rise rate of the upper surface **10sa** is 4° C./second or more. The temperature rise rate of the lower surface **10sb** is higher than the temperature rise rate of the upper surface **10sa**. The softening point T1 (° C.) of the first thermoplastic resin contained in the antioxidation film **12** is equal to or lower than the softening point T2 (° C.) of the second thermoplastic resin contained in the temporary fixing film **13**. That is, the softening temperature of the antioxidation film **12** is equal to or lower than the softening temperature of the temporary fixing film **13**. In the example shown in FIG. 2, a cooling step is performed in a cooling zone after the reflow step is finished in a heating zone.

(66) The manufacturing method (M) according to the present invention may include the step (x) of forming the plurality of solder precoats **11** and the antioxidation film **12** on the plurality of lands **10b**. FIGS. 3A to 3E schematically show an example of the step (x). First, the printed circuit board (substrate) **10** shown in FIG. 3A is prepared. The printed circuit board **10** includes the plate-shaped substrate portion **10a** and the lands **10b** formed on the surface of the substrate portion **10a**. The lands **10b** are connected to interconnects (not shown) on the substrate portion **10a** as necessary.

(67) Next, a mask **20** is placed on the printed circuit board (substrate) **10** as shown in FIG. 3B. The mask **20** includes openings **20h** at positions corresponding to the lands **10b** to which a solder paste is to be applied. Next, a solder paste **11p** is placed on the lands **10b** by being applied to the lands **10b** as shown in FIG. 3C. The solder paste **11p** contains solder particles **11a** and a flux **11f**.

(68) Next, the mask **20** is removed from the printed circuit board **10** as shown in FIG. 3D. Next, the printed circuit board **10** including the solder paste **11p** placed thereon is heated to melt the solder particles **11a**, and then the printed circuit board **10** is cooled. Thus, solder precoats **11** are formed on the lands **10b** as shown in FIG. 3E. At this time, a residue of the flux **11f** forms the antioxidation film **12**. A thermoplastic resin contained in the flux **11f** serves as the first thermoplastic resin contained in the antioxidation film **12**.

(69) Conventionally, it is thought that when a residue of the flux **11f** is left, the residue causes the occurrence of soldering defects in the reflow step. Therefore, the remaining flux residue has been conventionally removed by cleaning. However, in the present embodiment, the flux residue is used as the antioxidation film **12** on the solder precoats **11**. The antioxidation film **12** remains until just before the solder melts in the reflow step, and therefore, oxidation of the surfaces of the solder precoats **11** can be suppressed. Moreover, when the flux residue is used as the antioxidation film **12**, it is possible to omit a step of removing the flux residue.

(70) The manufacturing method (M) may include the step (y) and/or the step (z) described above, after the step (x) and before the step (i). FIGS. 4A and 4B show cross-sectional views of a step in an example of the case where the step (y) and the step (z) are performed after the step (x).

(71) The step (z) is performed in a case where at least one solder precoat **11** has been formed inappropriately in such a manner as to short-circuit two adjacent lands **10b** as shown in FIG. 4A. Note that, in the case where the step (z) is performed, an inspection step of inspecting whether or not solder precoats **11** have been formed appropriately is performed after the step (x) and before the step (z). The inspection step can be performed using a known method. For example, it is possible to detect a defect in solder precoats **11** by capturing an image of the printed circuit board **10** on which the solder precoats **11** have been formed, with a camera, and performing image processing on the captured image.

(72) In the step (y), the plurality of solder precoats **11** and the antioxidation film **12** are pressed from above the printed circuit board **10** with use of a flattening tool **51** that has a flat pressing surface as shown in FIG. 4B (flattening step). Thus, cracks (not shown) are formed in the antioxidation film **12**, upper surfaces **11t** of the solder precoats are flattened, and thicknesses of the solder precoats **11** are reduced. When the electronic components **30** are downsized, the solder precoats **11** are also downsized. When the solder precoats **11** are downsized, the upper surfaces **11t** of the solder precoats **11** may have hemispherical shapes under the influence of surface tension, and

it may be difficult to stably dispose the electronic components **30**. When the upper surfaces of the solder precoat **11** are flattened in the step (y), the electronic components **30** can be stably disposed. Also, when cracks are formed in the antioxidation film **12**, mixing of a solvent contained in the temporary fixing film **13** and a solvent contained in the antioxidation film **12** is facilitated. Also, the antioxidation film **12** peels off and anew surface of solder metal appears, and therefore, connection between electrodes of the electronic component and the solder constituting the solder precoat is facilitated. The antioxidation film **12** is supplied immediately after the flattening step, and therefore, the new surface of solder metal is hardly oxidized.

(73) Next, the step (z) is performed. First, as shown in FIG. 4C, a notch **11k** is formed in a portion of the inappropriately formed solder precoat **11**, the portion being between and above the two lands **10b**. The notch **11k** can be formed by pricking the above-described portion of the solder precoat **11** with a tool **52** like a chisel as shown in FIG. 4C, for example.

(74) Thereafter, the electronic components **30** are mounted in the same manner as that shown in FIGS. 1A to 1G. Specifically, first, the temporary fixing film **13** is formed in such a manner as to cover the solder precoat **11** and the antioxidation film **12** as shown in FIG. 4D (step (i)).

(75) Next, the plurality of electronic components **30** are disposed on the plurality of solder precoat **11** via the antioxidation film **12** and the temporary fixing film **13** as shown in FIG. 4E (step (ii)). Thus, the substrate **10X** on which the plurality of electronic components **30** are disposed is obtained.

(76) Next, the substrate **10X** is heated to melt the solder precoat **11** and then the molten solder is solidified, and thus the electronic components **30** are mounted on the printed circuit board **10** (step (iii)). This step is schematically shown in the cross-sectional views of FIGS. 4F to 4H. FIG. 4F shows a state where the antioxidation film **12** has softened and spread. FIG. 4G shows a state where the temporary fixing film **13** has softened and the components of the antioxidation film **12** and the temporary fixing film **13** are mixed with each other to be a mixture **13a**. Also, portions of the solder precoat **11** have melted and formed molten cores **11b**. FIG. 4H shows a state where the molten solder has been solidified and formed solder portions **11d**. The mixture **13a** has become a residue **13g**. Thus, the substrate **10Y** on which the electronic components **30** are mounted is obtained. At the time when the solder precoat **11** melt, portions of the solder precoat **11** that have not melted are attracted toward the molten cores **11b**, which have already melted, while melting. Therefore, the short circuit is finally solved as shown in FIG. 4H. When the notch **11k** is formed, the solder precoat **11** between the two lands **10b** is more likely to be attracted toward the molten cores **11b**. Therefore, it becomes easier to solve the short circuit between the two lands.

EXAMPLES

(77) The following describes the present invention more specifically using examples. In the following examples, electronic components were mounted with components of the antioxidation film and components of the temporary fixing film changed.

Experimental Example 1

(78) In Experimental Example 1, solder precoat were formed on lands on a printed circuit board with use of a solder paste. A film of a flux residue formed on surfaces of the solder precoat was used as an antioxidation film. A mixture containing solder particles (90 mass %), a polymerized rosin (4.5 mass %, softening point: 140° C.), abietic acid (0.5 mass %), castor wax (0.5 mass %), and hexyl diglycol (4.5 mass %) was used as the solder paste. The hexyl diglycol is a liquid component (a solvent), the abietic acid is an activating agent, and the castor wax is a thixotropic agent (the same also applies to the following examples). Particles of solder, which was a SnAgCu alloy, were used as the solder particles. The main component of the antioxidation film (the film of the flux residue) formed using the solder paste is the polymerized rosin, and it can be considered that the antioxidation film starts to soften at 140° C.

(79) Next, a temporary fixing film was formed in such a manner as to cover the solder precoat and the antioxidation film. A mixture containing a hydrogenated rosin (45 mass %, softening point: 75°

C.), abietic acid (5 mass %), caster wax (5 mass %), and hexyl diglycol (45 mass %) was used as the material of the temporary fixing film. The main component of the temporary fixing film is the hydrogenated rosin, and it can be considered that the temporary fixing film starts to soften at 75° C. (80) 20 electronic components were disposed on the solder precoat on the obtained substrate via the antioxidation film and the temporary fixing film. Electronic components having the 0201 size (a planar shape having a size of 0.25 mm×0.125 mm) specified in JIS were used as the electronic components. FIG. 5 is a top view schematically showing arrangement of the lands and the electronic components on the printed circuit board. In FIG. 5, the lands are hatched, and dotted lines show positions at which the electronic components were disposed. FIG. 5 also shows sizes of the lands and the electronic components (unit: mm). As shown in FIG. 5, the shortest distance L between adjacent lands was 0.05 mm.

(81) Next, the solder precoat was melted in a reflow apparatus to mount the electronic components on the printed circuit board. Thus, an electronic-component-mounted substrate of Experimental Example 1 was produced.

Experimental Examples 2 to 5

(82) Solder pastes and materials of the temporary fixing film were prepared by mixing the components at the same ratios as those adopted in Experimental Example 1, with the polymerized rosin contained in the solder paste and/or the hydrogenated rosin contained in the temporary fixing film changed to other rosins. Electronic-component-mounted substrates of Experimental Examples 2 to 5 were produced in the same manner as in Experimental Example 1 with use of the prepared solder pastes and materials.

(83) Whether or not the 20 electronic components were appropriately soldered to the lands was inspected for the electronic-component-mounted substrates of Experimental Examples 1 to 5. Specifically, the inspection was performed as to whether or not there was a defect (bridging), i.e., an electronic component being in contact with a land that was not a land to which the electronic component should have been soldered but was a land adjacent to that land. Table 1 shows rosins used in Experimental Examples 1 to 5, softening points of the rosins, and inspection results. Defect percentages shown in Table 1 indicate percentages of electronic components for which bridging occurred, out of the 20 electronic components.

(84) TABLE-US-00001

TABLE 1	Main component of antioxidation film	temporary fixing film	Softening point (° C.)	Softening point (° C.)	Defect percentage (%)
example 1	Polymerized rosin	Hydrogenated rosin	140	75	25
example 2	Polymerized rosin	Acrylated rosin	140	128	15
example 3	Polymerized rosin	Hydrogenated rosin	140	75	0
example 4	Polymerized rosin	Acrylated rosin	140	128	0
example 5	Polymerized rosin	Hydrogenated rosin	140	75	0

(85) As shown in Table 1, bridging did not occur in Experimental Examples 3 to 5, in which the softening point T1 was equal to or lower than the softening point T2. On the other hand, bridging occurred in Experimental Examples 1 and 2, in which the softening point T1 was higher than the softening point T2.

INDUSTRIAL APPLICABILITY

(86) The present invention is applicable to a method for manufacturing an electronic-component-mounted substrate.

(87) Although the present invention has been described in terms of the presently preferred embodiments, such a disclosure is not to be interpreted as limiting. Various alterations and modifications will no doubt become apparent to those skilled in the art to which the present invention pertains, after having read the above disclosure. Accordingly, it is intended that the appended claims be interpreted as covering all alterations and modifications that fall within the true spirit and scope of the present invention.

REFERENCE SIGNS LIST

(88) **10**: printed circuit board (substrate) **10X**, **10Y**: substrate **10b**: land **10sa**: upper surface **10sb**:

lower surface **11**: solder precoat **11a**: solder particle **11c**, **11d**: solder **11f** flux **11k**: notch **11p**: solder paste **11t**: upper surface **12**: antioxidation film **13**: temporary fixing film **30**: electronic component

Claims

1. A method for manufacturing an electronic-component-mounted substrate, the method comprising: a step (i) of forming a temporary fixing film in such a manner as to cover a plurality of solder precoat formed on a plurality of lands on a substrate and an antioxidation film formed in such a manner as to cover each of the plurality of solder precoat; a step (ii) of disposing a plurality of electronic components on the plurality of solder precoat via the antioxidation film and the temporary fixing film; and a step (iii) of soldering the plurality of electronic components to the plurality of lands by melting the plurality of solder precoat, wherein the antioxidation film contains a first thermoplastic resin, the temporary fixing film contains an activating agent and a second thermoplastic resin, a softening point of the first thermoplastic resin is lower than a softening point of the second thermoplastic resin, the method further comprising, before the step (i), a step (x) of forming the plurality of solder precoat and the antioxidation film on the plurality of lands, the antioxidation film is a residue formed on a surface of the solder precoat when the plurality of solder precoat are formed in the step (x).
 2. The method according to claim 1, wherein a shortest distance between a first land on which one electronic component out of the plurality of electronic components is disposed and a second land on which an electronic component adjacent to the one electronic component is disposed is 0.4 mm or less.
 3. The method according to claim 1, further comprising, after the step (x) and before the step (i), a step (y) of pressing the plurality of solder precoat and the antioxidation film from above the substrate to form cracks in the antioxidation film and flatten upper surfaces of the plurality of solder precoat.
 4. The method according to claim 1, further comprising, after the step (x) and before the step (i), a step (z) of forming a notch in a portion of at least one of the solder precoat formed in such a manner as to short-circuit two adjacent lands in the step (x), the portion being between and above the two adjacent lands.
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